C

ACT SUBJECT TEST — MATH 2 • UNIT 6.01

Limits, Asymptotes & Instantaneous Rates

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 A	Q2 A	Q3 A	Q4 A	Q5 B
Q6 A	Q7 A	Q8 A	Q9 C	Q10 B
Q11 A	Q12 B	Q13 B	Q14 A	Q15 B
Q16 D	Q17 A	Q18 A	Q19 C	Q20 C
Q21 A	Q22 B	Q23 D	Q24 A	Q25 A

Limits, Asymptotes & Instantaneous Rates Practice Exercises

1 MCQ

The graph of the function $f(x) = (x-1)^2 - 4$ is shown in the standard (x, y) coordinate plane below. What is the range of f?

A $[-4, \infty)$ ✓

B $[1, \infty)$

C $(-\infty, -4]$

D $(-\infty, \infty)$

WORKED SOLUTION & VERIFICATION

1. **Identify the vertex and direction of opening**: The function $f(x) = (x-1)^2 - 4$ is written in vertex form, $f(x) = a(x-h)^2 + k$, where $a = 1$, $h = 1$, and $k = -4$. The vertex of the parabola is at $(1, -4)$.
2. **Determine the minimum output value**: Since the leading coefficient $a = 1 > 0$, the parabola opens upward, meaning that the vertex represents the absolute minimum point on the graph. The minimum value attained by $f(x)$ is -4.
3. **Determine the range**: The range represents all possible output values (y-values) of the function. Because the function takes on its minimum value of -4 at $x = 1$ and increases without bound as x moves away from 1, the range is all real numbers greater than or equal to -4, which is written in interval notation as $[-4, \infty)$.

Common Misconceptions:

- Choice B, $[1, \infty)$, incorrectly uses the x-coordinate of the vertex instead of the y-coordinate.
- Choice C, $(-\infty, -4]$, incorrectly assumes the parabola opens downward.
- Choice D, $(-\infty, \infty)$, confuses the range (the set of y-values) with the domain (the set of all real x-values).

2 MCQ

For the complex number $i = \sqrt{-1}$, which of the following is equivalent to the expression $(5 - 3i) - (2 - 7i)$?

- **A** $3 + 4i$ ✓
- **B** $3 - 10i$
- **C** $7 + 4i$
- **D** $3 - 4i$

WORKED SOLUTION & VERIFICATION

To subtract complex numbers, distribute the negative sign across the second term and combine like terms (real parts with real parts, and imaginary parts with imaginary parts):

$(5 - 3i) - (2 - 7i) = 5 - 3i - 2 + 7i$

Group the real and imaginary components:

- Real part: $5 - 2 = 3$
- Imaginary part: $-3i + 7i = 4i$

Combining these gives $3 + 4i$.

Common misconception checks:

- Choice B $(3 - 10i)$ occurs if the negative sign is not distributed to the imaginary term, mistakenly calculating $-3i - 7i = -10i$.
- Choice C $(7 + 4i)$ results from accidentally adding the real parts $(5 + 2 = 7)$ while subtracting the imaginary parts.
- Choice D $(3 - 4i)$ results from a sign error when computing $-3 - (-7)$.

3 MCQ

What is the value of $\lim_{x \to 3} \frac{x^2 - 2x - 3}{x - 3}$?

- **A** 4 ✓
- **B** 0
- **C** 6
- **D** The limit does not exist

WORKED SOLUTION & VERIFICATION

Direct substitution of $x = 3$ into the expression gives $\frac{3^2 - 2(3) - 3}{3 - 3} = \frac{0}{0}$, which is an indeterminate form.

Factor the numerator:

$$x^2 - 2x - 3 = (x - 3)(x + 1)$$

For all $x \neq 3$, simplify the rational expression by canceling the common factor $(x - 3)$:

$$\frac{(x - 3)(x + 1)}{x - 3} = x + 1$$

Now evaluate the limit by direct substitution:

$$\lim_{x \to 3} (x + 1) = 3 + 1 = 4$$

Common misconceptions include concluding that the limit is 0 because the numerator evaluates to 0, or incorrectly believing that the limit does not exist simply because the denominator evaluates to 0 at $x = 3$.

4 MCQ

Given the vectors $\mathbf{u} = \langle 3, -4 \rangle$ and $\mathbf{v} = \langle -2, 5 \rangle$, what is the vector $2\mathbf{u} + \mathbf{v}$?

- A $\langle 4, -3 \rangle$ ✓
- B $\langle 1, 1 \rangle$
- C $\langle 4, 1 \rangle$
- D $\langle 8, -13 \rangle$

WORKED SOLUTION & VERIFICATION

To find $2\mathbf{u} + \mathbf{v}$, first perform the scalar multiplication on $\mathbf{u}$:

$$2\mathbf{u} = 2\langle 3, -4 \rangle = \langle 2(3), 2(-4) \rangle = \langle 6, -8 \rangle$$

Next, add the corresponding components of $2\mathbf{u}$ and $\mathbf{v}$:

$$2\mathbf{u} + \mathbf{v} = \langle 6 + (-2), -8 + 5 \rangle = \langle 4, -3 \rangle$$

- $\langle 1, 1 \rangle$ is incorrect because it represents $\mathbf{u} + \mathbf{v}$, omitting the scalar factor of 2.
- $\langle 4, 1 \rangle$ is incorrect because the scalar 2 was only distributed to the horizontal component.
- $\langle 8, -13 \rangle$ is incorrect because it calculates $2\mathbf{u} - \mathbf{v}$ instead of $2\mathbf{u} + \mathbf{v}$.

5 MCQ

The first four terms of an arithmetic sequence are 5, 12, 19, and 26. What is the 20th term of this sequence?

- A 131
- B **138** ✓
- C 140
- D 145

WORKED SOLUTION & VERIFICATION

To find the 20th term of the arithmetic sequence, first identify the initial term and the common difference:

1. The first term is $a_1 = 5$.
2. The common difference d is found by subtracting consecutive terms: $d = 12 - 5 = 7$.

The general formula for the nth term of an arithmetic sequence is:

$$a_n = a_1 + (n - 1)d$$

Substitute $n = 20$, $a_1 = 5$, and $d = 7$:

$$a_{20} = 5 + (20 - 1)(7) = 5 + 19(7) = 5 + 133 = 138$$

Common misconceptions:

- Using n instead of $n - 1$ yields $5 + 20(7) = 145$.
- Calculating $20 \times 7 = 140$ overlooks the starting value $a_1 = 5$.
- Using $n - 2$ steps yields $5 + 18(7) = 131$.

6 MCQ

A point has polar coordinates $(r, \theta) = (6, \frac{2\pi}{3})$. What are the Cartesian coordinates (x, y) of this point?

- **A** $(-3, 3\sqrt{3})$ ✓
- **B** $(-3\sqrt{3}, 3)$
- **C** $(3, -3\sqrt{3})$
- **D** $(3\sqrt{3}, -3)$

WORKED SOLUTION & VERIFICATION

To convert polar coordinates (r, θ) to Cartesian coordinates (x, y), use the conversion formulas:

$$x = r\cos\theta \quad \text{and} \quad y = r\sin\theta$$

Given $r = 6$ and $\theta = \frac{2\pi}{3}$:

$$x = 6\cos\left(\frac{2\pi}{3}\right) = 6\left(-\frac{1}{2}\right) = -3$$

$$y = 6\sin\left(\frac{2\pi}{3}\right) = 6\left(\frac{\sqrt{3}}{2}\right) = 3\sqrt{3}$$

Thus, the Cartesian coordinates are $(-3, 3\sqrt{3})$.

Common errors:

- Interchanging x and y (using $x = r\sin\theta$ and $y = r\cos\theta$) yields $(3\sqrt{3}, -3)$.
- Reversing the quadrant signs or applying $\frac{\pi}{6}$ reference angle formulas erroneously produces $(-3\sqrt{3}, 3)$.

7 MCQ

The polynomial function $P(x) = 2x^3 + ax^2 - 5x + b$ has $(x - 1)$ as a factor. When $P(x)$ is divided by $(x + 2)$, the remainder is -18. What is the remainder when $P(x)$ is divided by $(x - 3)$?

- **A** 2 ✓
- **B** 8
- **C** 18
- **D** -30

WORKED SOLUTION & VERIFICATION

According to the Factor Theorem and the Remainder Theorem, if a polynomial $P(x)$ is divided by $(x - c)$, the remainder is $P(c)$. If $(x - c)$ is a factor, then $P(c) = 0$.

Step 1: Set up the first equation using the factor $(x - 1)$:

$$P(1) = 2(1)^3 + a(1)^2 - 5(1) + b = 0$$

$$2 + a - 5 + b = 0 \implies a + b = 3$$

Step 2: Set up the second equation using the divisor $(x + 2) = (x - (-2))$ with remainder -18:

$$P(-2) = 2(-2)^3 + a(-2)^2 - 5(-2) + b = -18$$

$$2(-8) + 4a + 10 + b = -18$$

$$-16 + 10 + 4a + b = -18 \implies 4a + b = -12$$

Step 3: Solve the system of linear equations for a and b:

Subtracting the first equation from the second:

$$(4a + b) - (a + b) = -12 - 3$$

$$3a = -15 \implies a = -5$$

Substitute $a = -5$ into $a + b = 3$:

$$-5 + b = 3 \implies b = 8$$

Thus, the polynomial is $P(x) = 2x^3 - 5x^2 - 5x + 8$.

Step 4: Find the remainder when $P(x)$ is divided by $(x - 3)$ by evaluating $P(3)$:

$$P(3) = 2(3)^3 - 5(3)^2 - 5(3) + 8 = 2(27) - 5(9) - 15 + 8 = 54 - 45 - 15 + 8 = 2$$

Common Pitfalls:

- Evaluating $P(2)$ instead of $P(-2)$ produces $4a + b = -24$, leading to $a = -9$, $b = 12$, and an incorrect remainder of -30.
- Confusing the constant term $b = 8$ with the final answer yields 8.

8 MCQ

For what real value of k is the complex expression

$$\frac{k+8i}{3-i}+\frac{2-ki}{3+i}$$

a real number?

- **A** 11 ✓
- **B** 1
- **C** 6
- **D** -11

WORKED SOLUTION & VERIFICATION

To determine when the sum of the two fractions is a real number, simplify the expression by combining terms over a common denominator or multiplying by conjugates.

The common denominator of the two fractions is:

$$(3-i)(3+i)=3^2-i^2=9-(-1)=10$$

Rewrite each term over this denominator:

1. For the first term:

$$\frac{(k+8i)(3+i)}{(3-i)(3+i)}=\frac{3k+ki+24i+8i^2}{10}=\frac{(3k-8)+(k+24)i}{10}$$

2. For the second term:

$$\frac{(2-ki)(3-i)}{(3+i)(3-i)}=\frac{6-2i-3ki+ki^2}{10}=\frac{(6-k)+(-3k-2)i}{10}$$

Now, sum the real and imaginary components:

$$\text{Real part}=\frac{(3k-8)+(6-k)}{10}=\frac{2k-2}{10}$$

$$\text{Imaginary part}=\frac{(k+24)+(-3k-2)}{10}=\frac{-2k+22}{10}$$

For the resulting complex number to be a real number, its imaginary part must equal 0:

$$\frac{-2k+22}{10}=0 \implies -2k+22=0 \implies 2k=22 \implies k=11$$

Common Misconceptions:

- Choosing **1** results from setting the real part equal to 0 $(2k-2=0)$, which would make the expression purely imaginary rather than real.
- Choosing **6** results from equating the real and imaginary parts $(2k-2=-2k+22)$.
- Choosing **-11** results from an algebraic sign error when solving $-2k=-22$.

9 MCQ

For real constants a and b, suppose that

$$\lim_{x \to 0} \frac{\sqrt{ax+b}-3}{x} = 2.$$

What is the value of $a+b$?

A 12

B 15

C 21 ✓

D 27

WORKED SOLUTION & VERIFICATION

Step 1: Determine the value of b.

As $x \to 0$, the denominator approaches 0. For the limit to exist and equal the finite real number 2, the numerator must also approach 0 to create an indeterminate form of $\frac{0}{0}$:

$$\lim_{x \to 0} (\sqrt{ax+b}-3) = \sqrt{b}-3 = 0 \implies \sqrt{b} = 3 \implies b = 9.$$

Step 2: Evaluate the limit in terms of a.

Substitute $b = 9$ and rationalize the numerator by multiplying by the conjugate:

$$\lim_{x \to 0} \frac{\sqrt{ax+9}-3}{x} \cdot \frac{\sqrt{ax+9}+3}{\sqrt{ax+9}+3} = \lim_{x \to 0} \frac{(ax+9)-9}{x(\sqrt{ax+9}+3)}$$

$$= \lim_{x \to 0} \frac{ax}{x(\sqrt{ax+9}+3)} = \lim_{x \to 0} \frac{a}{\sqrt{ax+9}+3}$$

Evaluating the limit at $x = 0$:

$$\frac{a}{\sqrt{9}+3} = \frac{a}{6}.$$

Step 3: Solve for a and find $a+b$.

We are given that the limit equals 2:

$$\frac{a}{6} = 2 \implies a = 12.$$

Thus, $a+b = 12+9 = 21$.

Common Misconceptions:

- Choice A (12): Solves correctly for a but forgets to add b.
- Choice B (15): Omits the factor of 2 in the denominator when differentiating or rationalizing, leading to $\frac{a}{3} = 2 \implies a = 6$, giving $6+9 = 15$.
- Choice D (27): Incorrectly squares 9 or miscalculates the conjugate sum as 3.

10 MCQ

Let $P(x)$ be a polynomial of minimum degree with real coefficients such that

$$\lim_{x\to 1}\frac{P(x)}{x^2-3x+2}=5 \quad \text{and} \quad \lim_{x\to 2}\frac{P(x)}{x^2-3x+2}=-3.$$

What is the value of $P(3)$?

A -11

B **-22** ✓

C 22

D 42

WORKED SOLUTION & VERIFICATION

First, factor the denominator: $x^2-3x+2=(x-1)(x-2)$.

For $\lim_{x\to 1}\frac{P(x)}{(x-1)(x-2)}$ to exist and be finite when the denominator approaches 0, the numerator must also approach 0. Thus, $P(1)=0$, meaning $(x-1)$ is a factor of $P(x)$.

Similarly, for $\lim_{x\to 2}\frac{P(x)}{(x-1)(x-2)}$ to exist, we must have $P(2)=0$, so $(x-2)$ is also a factor of $P(x)$.

Therefore, $P(x)$ must be divisible by $(x-1)(x-2)$. We can write:

$$P(x)=(x-1)(x-2)Q(x)$$

for some polynomial $Q(x)$.

For all $x\neq 1,2$, the expression simplifies to $Q(x)$. Since polynomials are continuous everywhere:

$$\lim_{x\to 1}Q(x)=Q(1)=5$$

$$\lim_{x\to 2}Q(x)=Q(2)=-3$$

If $Q(x)$ were a constant, $Q(1)$ and $Q(2)$ would be equal, which is impossible since $5\neq -3$. Thus, the minimum degree of $Q(x)$ is 1. Let $Q(x)=ax+b$.

Setting up the system of linear equations:

1) $a(1)+b=5 \implies a+b=5$

2) $a(2)+b=-3 \implies 2a+b=-3$

Subtracting the first equation from the second gives:

$$a=-3-5=-8$$

Then:

$$b=5-(-8)=13$$

So, $Q(x)=-8x+13$, and the polynomial of minimum degree is:

$$P(x)=(x-1)(x-2)(-8x+13)$$

Now evaluate $P(3)$:

$$P(3)=(3-1)(3-2)(-8(3)+13)=(2)(1)(-24+13)=2(-11)=-22.$$

Common Misconceptions:

- Forgetting to multiply by the factor $(x-1)(x-2)=2$ yields $Q(3)=-11$.
- A sign error when solving for the slope (e.g., $a=8, b=-3$) yields $P(3)=42$.
- A sign error when evaluating $-24+13$ yields 22.

11 MCQ

What is the value of the limit $\lim_{x \to 2} \frac{x^2+x-6}{x-2}$?

- **A** 5 ✓
- **B** 1
- **C** -1
- **D** Does not exist

WORKED SOLUTION & VERIFICATION

Direct substitution of $x = 2$ gives the indeterminate form:

$$\frac{2^2 + 2 - 6}{2 - 2} = \frac{0}{0}$$

To evaluate the limit, factor the quadratic expression in the numerator:

$$x^2 + x - 6 = (x - 2)(x + 3)$$

For all $x \neq 2$, we can cancel the common factor $(x - 2)$:

$$\frac{x^2 + x - 6}{x - 2} = \frac{(x - 2)(x + 3)}{x - 2} = x + 3$$

Now, evaluate the limit by direct substitution:

$$\lim_{x \to 2}(x + 3) = 2 + 3 = 5$$

Common misconceptions include:
- Factoring incorrectly as $(x - 2)(x - 3)$, leading to $2 - 3 = -1$.
- Assuming the limit does not exist simply because the denominator equals 0 upon direct substitution, overlooking that the numerator is also 0 (removable discontinuity).

12 MCQ

If the limit $\lim_{x\to 3} \frac{x^2+kx-15}{x^2-4x+3} = L$ exists and is a finite real number, where k and L are constants, what is the value of $k + L$?

A 4

B 6 ✓

C 8

D 2

WORKED SOLUTION & VERIFICATION

First, evaluate the limit of the denominator as $x \to 3$:

$$3^2 - 4(3) + 3 = 9 - 12 + 3 = 0$$

Since the denominator approaches 0 and the limit L is a finite real number, the numerator must also approach 0 as $x \to 3$ to produce an indeterminate form of $\frac{0}{0}$. Setting the numerator equal to 0 at $x = 3$:

$$3^2 + 3k - 15 = 0 \implies 9 + 3k - 15 = 0 \implies 3k - 6 = 0 \implies k = 2$$

Next, substitute $k = 2$ back into the rational function and factor both numerator and denominator:

$$\frac{x^2 + 2x - 15}{x^2 - 4x + 3} = \frac{(x-3)(x+5)}{(x-3)(x-1)}$$

For all $x \neq 3$, cancel the common factor $(x - 3)$:

$$L = \lim_{x\to 3} \frac{x+5}{x-1} = \frac{3+5}{3-1} = \frac{8}{2} = 4$$

Finally, find $k + L$:

$$k + L = 2 + 4 = 6$$

Common traps:
- Finding only $k = 2$ yields choice A.
- Finding only $L = 4$ yields choice B.
- Multiplying $k \cdot L = 2 \cdot 4 = 8$ instead of adding yields choice D.

13 MCQ

For each real number $t \in (-2, 2)$, the cubic equation $x^3 - 3x + t = 0$ has three distinct real roots, denoted in increasing order by $x_1(t) < x_2(t) < x_3(t)$.

What is the value of the limit

$$\lim_{t \to 2^-} \frac{x_3(t) - x_2(t)}{\sqrt{2 - t}} \, ?$$

A $\sqrt{3}$

B $\frac{2\sqrt{3}}{3}$ ✓

C $\frac{\sqrt{3}}{2}$

D $\frac{4}{3}$

WORKED SOLUTION & VERIFICATION

To evaluate the limit, we analyze the behavior of the roots as $t \to 2^-$ using polynomial factorization and Vieta's formulas.

Step 1: Determine the limiting values of the roots.

When $t = 2$, the equation becomes:

$$x^3 - 3x + 2 = (x - 1)^2(x + 2) = 0$$

The roots are -2 (simple root) and 1 (double root). Since $x_1(t) < x_2(t) < x_3(t)$, as $t \to 2^-$:

$$\lim_{t \to 2^-} x_1(t) = -2, \quad \lim_{t \to 2^-} x_2(t) = 1, \quad \lim_{t \to 2^-} x_3(t) = 1$$

Step 2: Relate $(x_3 - x_2)^2$ to x_1 via Vieta's formulas.

For the cubic $x^3 - 3x + t = 0$, the roots x_1, x_2, x_3 satisfy:

$$x_1 + x_2 + x_3 = 0 \implies x_2 + x_3 = -x_1$$

$$x_1(x_2 + x_3) + x_2x_3 = -3 \implies x_1(-x_1) + x_2x_3 = -3 \implies x_2x_3 = x_1^2 - 3$$

Now, compute the squared difference between the two larger roots:

$$(x_3 - x_2)^2 = (x_2 + x_3)^2 - 4x_2x_3 = (-x_1)^2 - 4(x_1^2 - 3) = 12 - 3x_1^2 = 3(4 - x_1^2) = 3(2 - x_1)(2 + x_1)$$

Step 3: Express $2 + x_1$ in terms of $2 - t$.

Since x_1 is a root of $x^3 - 3x + t = 0$, we have $t = 3x_1 - x_1^3$. Thus:

$$2 - t = 2 - 3x_1 + x_1^3 = (x_1 - 1)^2(x_1 + 2)$$

Rearranging gives:

$$x_1 + 2 = \frac{2 - t}{(x_1 - 1)^2}$$

Step 4: Evaluate the limit.

Substitute this expression into the squared difference:

$$(x_3 - x_2)^2 = 3(2 - x_1) \cdot \frac{2 - t}{(x_1 - 1)^2} \implies \frac{(x_3 - x_2)^2}{2 - t} = \frac{3(2 - x_1)}{(x_1 - 1)^2}$$

As $t \to 2^-$, $x_1 \to -2$. Taking the limit:

$$\lim_{t \to 2^-} \frac{(x_3 - x_2)^2}{2 - t} = \frac{3(2 - (-2))}{(-2 - 1)^2} = \frac{3 \cdot 4}{9} = \frac{4}{3}$$

Because $x_3(t) > x_2(t)$, the difference $x_3(t) - x_2(t)$ is strictly positive. Taking the square root gives:

$$\lim_{t\to 2^-}\frac{x_3(t)-x_2(t)}{\sqrt{2-t}}=\sqrt{\frac{4}{3}}=\frac{2}{\sqrt{3}}=\frac{2\sqrt{3}}{3}$$

Misconception Alert:
- Choice D ($\frac{4}{3}$) arises from forgetting to take the square root after computing the limit of the squared ratio.
- Choice B ($\sqrt{3}$) results from incorrectly approximating $(x_3-x_2)^2\approx 3(2-t)$.

14 MCQ

What is the value of $\lim_{x\to 2}\frac{x^2+x-6}{x-2}$?

A 5 ✓

B 1

C 0

D Does not exist

WORKED SOLUTION & VERIFICATION

Direct substitution of $x=2$ into the expression gives the indeterminate form:

$$\frac{2^2+2-6}{2-2}=\frac{0}{0}$$

To evaluate the limit, factor the quadratic numerator:

$$x^2+x-6=(x-2)(x+3)$$

For all $x\neq 2$, we can cancel the common factor $(x-2)$:

$$\frac{x^2+x-6}{x-2}=\frac{(x-2)(x+3)}{x-2}=x+3$$

Now, evaluate the limit by substituting $x=2$:

$$\lim_{x\to 2}(x+3)=2+3=5$$

Common Misconceptions:
- Incorrectly concluding the limit is 0 or undefined immediately upon encountering the indeterminate form $\frac{0}{0}$.
- Factoring errors such as writing $(x+2)(x-3)$, which leads to an erroneous value.

15 MCQ

For real numbers a and b, suppose that

$$\lim_{x \to 1} \frac{x^2 + ax + b}{x^2 - 1} = 3.$$

What is the value of $a - b$?

A -1

B 9 ✓

C 3

D -9

WORKED SOLUTION & VERIFICATION

For the rational function to have a finite limit as $x \to 1$ while the denominator $x^2 - 1 \to 0$, the numerator must also approach 0. Setting the numerator to 0 at $x = 1$:

$$1^2 + a(1) + b = 0 \implies 1 + a + b = 0 \implies b = -(a + 1).$$

Substitute $b = -(a + 1)$ back into the numerator:

$$x^2 + ax - (a + 1) = (x - 1)(x + a + 1).$$

Now evaluate the limit by canceling the common factor $(x - 1)$:

$$\lim_{x \to 1} \frac{(x - 1)(x + a + 1)}{(x - 1)(x + 1)} = \lim_{x \to 1} \frac{x + a + 1}{x + 1} = \frac{1 + a + 1}{1 + 1} = \frac{a + 2}{2}.$$

We are given that this limit equals 3:

$$\frac{a + 2}{2} = 3 \implies a + 2 = 6 \implies a = 4.$$

Then, find b:

$$b = -(4 + 1) = -5.$$

Finally, calculate $a - b$:

$$a - b = 4 - (-5) = 4 + 5 = 9.$$

16 MCQ

In the xy-plane, let $P(t, t^2)$ with $t > 0$ be a point on the parabola $y = x^2$. The normal line to the parabola at P intersects the parabola again at a second point Q. Let T be the intersection point of the tangent lines to the parabola at P and Q. As $t \to 0^+$, what is the limiting value of the y-coordinate of T?

A $-\frac{1}{4}$

B -1

C 0

D $-\frac{1}{2}$ ✓

WORKED SOLUTION & VERIFICATION

We synthesize differential calculus, coordinate geometry, and limits to determine the behavior of T as $t \to 0^+$.

1. **Equation of the Normal Line at P:**

Since $y = x^2$, the derivative is $y' = 2x$. At point $P(t, t^2)$, the slope of the tangent line is $2t$. Therefore, the slope of the normal line is the negative reciprocal:

$$m_{\text{normal}} = -\frac{1}{2t}$$

The equation of the normal line at P is:

$$y - t^2 = -\frac{1}{2t}(x - t)$$

2. **Finding the Second Intersection Point Q:**

To find where the normal line intersects $y = x^2$, substitute $y = x^2$:

$$x^2 - t^2 = -\frac{1}{2t}(x - t)$$

Factoring the left-hand side gives $(x - t)(x + t) = -\frac{1}{2t}(x - t)$. Since $Q \neq P$, we have $x \neq t$, so we divide by $x - t$:

$$x + t = -\frac{1}{2t} \implies x_Q = -t - \frac{1}{2t}$$

3. **Intersection T of the Tangent Lines:**

The tangent lines to $y = x^2$ at $x = x_P$ and $x = x_Q$ are:

$$y = 2tx - t^2$$

$$y = 2x_Q x - x_Q^2$$

Subtracting the two equations to find their intersection x_T:

$$2(t - x_Q)x_T = t^2 - x_Q^2 = (t - x_Q)(t + x_Q) \implies x_T = \frac{t + x_Q}{2}$$

Substitute $x_Q = -t - \frac{1}{2t}$:

$$x_T = \frac{t + (-t - \frac{1}{2t})}{2} = -\frac{1}{4t}$$

Now substitute x_T into the tangent line at P to find y_T:

$$y_T = 2t(-\frac{1}{4t}) - t^2 = -\frac{1}{2} - t^2$$

4. **Evaluating the Limit:**

Taking the limit as $t \to 0^+$:

$$\lim_{t\to 0^+} y_T = \lim_{t\to 0^+} \left(-\frac{1}{2} - t^2\right) = -\frac{1}{2}$$

Common misconceptions include assuming T approaches the origin $(0, 0)$ because $P \to (0, 0)$ (giving 0), or confusing the y-coordinate with the focus of the parabola $y = 1/4$ (giving $-1/4$).

17 MCQ

What is the value of $\lim_{x\to 3} \frac{2x^2-5x-3}{x-3}$?

- **A** 7 ✓
- **B** 5
- **C** 1
- **D** 0

WORKED SOLUTION & VERIFICATION

Direct substitution of $x = 3$ yields the indeterminate form $\frac{2(3)^2-5(3)-3}{3-3} = \frac{0}{0}$.
To evaluate the limit, factor the quadratic expression in the numerator:

$$2x^2 - 5x - 3 = (2x + 1)(x - 3)$$

For all $x \neq 3$, simplify the rational expression by canceling the common factor $(x - 3)$:

$$\frac{2x^2 - 5x - 3}{x - 3} = \frac{(2x + 1)(x - 3)}{x - 3} = 2x + 1$$

Now, evaluate the limit by direct substitution into the simplified expression:

$$\lim_{x\to 3}(2x + 1) = 2(3) + 1 = 7$$

Common misconceptions:
- Incorrectly concluding that a $\frac{0}{0}$ result equals 0.
- Factoring errors, such as using $(2x - 1)(x - 3)$, which yields $2(3) - 1 = 5$.

18 MCQ

Find $\lim_{x\to 4} (-3x^2 - 6)$.

- **A** **-54** ✓
- **B** -53
- **C** -56
- **D** 54

WORKED SOLUTION & VERIFICATION

Direct substitution (polynomials are continuous everywhere): plug in $x = 4$ to get -54.

19 MCQ

Find $\lim_{x\to -3} (-3x^2 + 4x + 3)$.

- **A** -35
- **B** -38
- **C** **-36** ✓
- **D** 36

WORKED SOLUTION & VERIFICATION

Direct substitution (polynomials are continuous everywhere): plug in $x = -3$ to get -36.

20 MCQ

Find $\lim_{x \to -3} (x^2 - 4x)$.

A 22	B 19
C **21** ✓	D -21

WORKED SOLUTION & VERIFICATION

Direct substitution (polynomials are continuous everywhere): plug in $x = -3$ to get 21.

21 MCQ

Find $\lim_{x \to 3} (-4x^2 + 4x + 4)$.

A **-20** ✓	B -19
C -22	D 20

WORKED SOLUTION & VERIFICATION

Direct substitution (polynomials are continuous everywhere): plug in $x = 3$ to get -20.

22 MCQ

Find $\lim_{x \to -2} (-3x^2 + 2x - 6)$.

A -21	B **-22** ✓
C -24	D 22

WORKED SOLUTION & VERIFICATION

Direct substitution (polynomials are continuous everywhere): plug in $x = -2$ to get -22.

23 MCQ

Find $\lim_{x \to 4} (-2x^2 - 3x - 4)$.

A -50	B 48
C -47	D **-48** ✓

WORKED SOLUTION & VERIFICATION

Direct substitution (polynomials are continuous everywhere): plug in $x = 4$ to get -48.

24 MCQ

Find $\lim_{x \to 0} (-2x^2 - 2x)$.

- **A** 0 ✓
- B 1
- C -2
- D 5

WORKED SOLUTION & VERIFICATION

Direct substitution (polynomials are continuous everywhere): plug in $x = 0$ to get 0.

25 MCQ

Find $\lim_{x \to 1} (4x^2 - 6x - 2)$.

- **A** -4 ✓
- B -3
- C -6
- D 4

WORKED SOLUTION & VERIFICATION

Direct substitution (polynomials are continuous everywhere): plug in $x = 1$ to get -4.